

Fractional Integral Calculus and Complex-Order Transform Hierarchies for the Fabius–Rvachev System

Fabius clocks, inverse-quantile transmutation, Weyl refinement,
Riesz tails, and generalized Stieltjes resolvents

Research report prepared for Vladimir Reshetnikov’s ProveIt development

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Abstract

Let F be the bounded Fabius distribution function, let $Q = F^{-1}$ on $(0, 1)$, and let up denote Rvachev’s even compactly supported density. The current ProveIt corpus already contains a substantial ordinary and Riemann–Liouville integration theory: exact integer primitive ladders, finite formulas for monomial weights, entire Newton–Mellin continuations, inverse-area identities, moment recurrences, sinc-product transforms, and a Lean formalization of the normalized integer Volterra ladder. This report uses those results as a baseline and develops a genuinely fractional and complex-order layer that was not found as a systematic subject in the recursively audited corpus.

The first new construction is an exact fractional calculus in the *Fabius clock*. Conjugating ordinary Riemann–Liouville and Caputo operators by the change of variable $y = F(x)$ gives

$$(I_F^\alpha h)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (F(x) - F(t))^{\alpha-1} h(t) F'(t) dt, \quad D_F = \frac{1}{F'} \frac{d}{dx}.$$

It has an exact semigroup, diagonalizes powers of F , and admits Mittag–Leffler eigenfunctions. The dual Q -clock calculus has the same properties. In ordinary time, a transmutation theorem gives, for $0 < \beta < 1$,

$${}^{\text{C}}D_{0+}^\beta [H(Q(y))] = \frac{1}{\Gamma(1-\beta)} \int_0^{Q(y)} (y - F(x))^{-\beta} H'(x) dx.$$

For $H(x) = x^r$ this yields a closed integral formula for the Caputo derivative of every positive power of the inverse Fabius function and the endpoint law

$${}^{\text{C}}D_{0+}^\beta Q(y)^r \sim \frac{Q(y)^r y^{-\beta}}{\Gamma(1-\beta)} \quad (y \downarrow 0).$$

A second exact bridge connects standard fractional derivatives of F to fractional primitives of up :

$$D_{0+}^\beta F(x) = 2^\beta (I_{-1+}^{1-\beta} \text{up})(2x - 1), \quad 0 < \beta < 1.$$

On the whole line, the Weyl family $W_\sigma = \mathcal{F}^{-1}[(i\xi)^\sigma \widehat{\text{up}}(\xi)]$ satisfies the complex-order refinement hierarchy

$$W_\sigma(x) = 2^\sigma (W_{\sigma-1}(2x + 1) - W_{\sigma-1}(2x - 1)).$$

This continuously interpolates Rvachev’s differential–difference equation.

The transform side is organized by the entire complex-order generalized Stieltjes function

$$S_\lambda(z) = \int_{-1}^1 \frac{\text{up}(t)}{(z - t)^\lambda} dt.$$

It obeys a dyadic order-lowering equation, an exact derivative hierarchy, and an even-moment series. It also controls nonlocal operators. For $0 < \beta < 2$ and $|x| > 1$,

$$(-\Delta)^{\beta/2} \text{up}(x) = -C_{1,\beta} S_{1+\beta}(|x|),$$

so the exterior fractional Laplacian has a full rational-moment expansion. A parallel formula is proved for Riesz potentials. Numerical Fourier experiments suggest that $(-\Delta)^{\beta/2}$ up has exactly one positive zero for every $0 < \beta < 2$; this is stated explicitly as a conjecture.

The report also develops complex-order resolvents containing F and Q , fractional autocorrelation energies of the infinite sinc product, a catalog of definite and indefinite integral identities, reproducible numerical checks, and a Lean formalization roadmap. “New” throughout means new relative to the pinned repository corpus after a targeted collision audit; it is not a claim of universal publication priority.

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1 Scope, recursive corpus audit, and proof status

1.1 Audit endpoint and moving-tree caveat

The requested documentation root was inspected recursively at the commit shown on the title page:

<https://github.com/VladimirReshetnikov/ProveIt/tree/3adc011108796355f1a874a5036f28098408d370/Analysis/FabiusFunction/docs>.

At the audit endpoint, the indexed recursive query returned 89 TeX paths. The branch was actively changing during the investigation: the final pinned merge added both a new integral-transform report and a Lean implementation of the normalized integer Volterra foundation. For that reason, every novelty comparison below is tied to the immutable commit 3adc011108796355f1a874a5036f28098408d370, not merely to the branch name `main`.

The audit was path-complete and content-deduplicated. Current expositions, frontier reports, source-paper transcriptions, the glossary, the Lean walkthrough, and mathematically distinctive archival versions were examined. Byte-identical or superseded historical copies were treated as provenance, not as independent mathematical sources. The principal families were:

- the live Fabius–Rvachev exposition, glossary, non-elementarity study, and Lean walkthrough;
- exact dyadic, q -binomial, Thue–Morse, Bell/Bernoulli, and Richardson-extrapolation reports;
- the inverse-Fabius and Lambert– W endpoint reports;
- repeated-integration, monomial-antiderivative, integral-transform, and derivative-norm reports;
- the complete Fourier-decay campaign for the infinite sinc product;
- archived predecessor documents and seven transcribed primary papers.

1.2 What the repository already contains

The following material is treated as *corpus baseline* rather than relabelled as a contribution of this report [9, 10, 11]:

- the probability model, functional–differential equations, signed Thue–Morse extension, partition of unity, and all integer derivatives;
- exact dyadic values, rational moment recurrences, q -Pochhammer and Gaussian-binomial formulas;
- sharp Fourier decay, zero multiplicities, lobe signs, and spectral determinant interpretations of the sinc product;
- Lambert– W endpoint asymptotics and their small periodic correction;
- inverse-function germs, Bell-polynomial inverse derivatives, and inverse endpoint asymptotics;
- normalized integer Volterra primitives, monomially weighted primitive formulas, and their recent Lean formalization;
- Riemann–Liouville fractional primitives, inverse-area formulas, Mellin/Newton moment continuations, beta weights, order statistics, integer resolvents, and several correlation integrals;
- integer derivative equimeasurability and all rearrangement-invariant norm identities.

Targeted searches were performed for *Caputo*, *fractional Laplacian*, *Riesz potential*, *Weyl derivative*, *fractional calculus with respect to F* , and the exact formula shapes proved below. The only Caputo occurrence located was a short remark noting that Caputo and Riemann–Liouville derivatives coincide for F because of endpoint flatness. No developed theory of the other structures was found.

1.3 Status ledger

Result family	Status	Main input
Integer derivative and primitive ladders; monomial Newton–Volterra series	corpus baseline	signed self-differential equation and endpoint flatness
Inverse-area identities and standard Riemann–Liouville primitives	corpus baseline	quantile substitution and Volterra integration
Fabius-clock and inverse-clock fractional calculi	proved here	exact conjugacy by $y = F(x)$ and $x = Q(y)$
Caputo/Riemann–Liouville transmutation for $H \circ Q$	proved here	inverse-chain operator $(F')^{-1}D$
Endpoint laws for fractional integrals and Caputo derivatives of Q^r	proved here	rapid variation and convexity of F near zero
Standard fractional bridge between F and up	proved here	$F^{(n)}(x) = 2^n \text{up}^{(n-1)}(2x - 1)$
Whole-line Weyl complex-order refinement hierarchy	proved here	Fourier multiplier calculus and the sinc refinement product
Generalized Stieltjes order hierarchy	proved here	random self-similarity $Z \stackrel{d}{=} (V + Z')/2$
Complex-order resolvents involving F , Q , and $1 - F$	proved here	integration by parts after quantile transport
Exterior Riesz fractional-Laplacian and potential expansions	proved here	compact support and the generalized Stieltjes series
One-node law for $(-\Delta)^{\beta/2} \text{up}$	conjectural	nonperiodic singular-integral profiles for seven fractional orders
Fractional endpoint phase correction and fractional-energy rigidity	conjectural	Lambert phase and integer spectral moment interpolation

Priority and regularity discipline

A theorem labelled “proved here” is supplied with a proof in this document, but has not necessarily been formalized in Lean. Novelty is relative to the pinned repository corpus. Standard constructions from fractional calculus are cited rather than claimed as new; the new content is their exact specialization and interaction with the Fabius–Rvachev structure. Whenever a Caputo derivative requires endpoint regularity that the inverse Q may not possess at higher order, the hypothesis is stated explicitly.

2 Normalization and structural input

2.1 The bounded function, inverse, and compact density

Following the probabilistic construction of Fabius and the compact-support formulation of Rvachev/Arias de Reyna [1, 2, 3], let $U_1, U_2, \dots$ be independent uniform random variables on $[0, 1]$ and set

$$Y = \sum_{j=1}^{\infty} 2^{-j} U_j, \quad Z = 2Y - 1 = \sum_{j=1}^{\infty} 2^{-j} V_j, \quad (2.1)$$

where $V_j = 2U_j - 1$ is uniform on $[-1, 1]$. The bounded Fabius function is the CDF of Y :

$$F(x) = \mathbb{P}(Y \leq x), \quad F(x) = 0 \ (x \leq 0), \quad F(x) = 1 \ (x \geq 1). \quad (2.2)$$

The variable Z has the even density up supported on $[-1, 1]$. Thus

$$F'(x) = 2 \operatorname{up}(2x - 1), \quad 0 < x < 1, \quad (2.3)$$

$$\operatorname{up}(t) = F(1 - |t|), \quad |t| \leq 1, \quad (2.4)$$

$$F(1 - x) = 1 - F(x), \quad Q(1 - y) = 1 - Q(y). \quad (2.5)$$

On $[0, 1]$, $\operatorname{up}(x) = 1 - F(x) = F(1 - x)$. The inverse $Q = F^{-1}$ is continuous on $[0, 1]$, smooth on $(0, 1)$, and satisfies

$$Q'(y) = \frac{1}{F'(Q(y))} = \frac{1}{2 \operatorname{up}(2Q(y) - 1)}. \quad (2.6)$$

The left-half self-differential equation is

$$F'(x) = 2F(2x), \quad 0 \leq x \leq \frac{1}{2}, \quad (2.7)$$

and Rvachev's global differential–difference equation is

$$\operatorname{up}'(x) = 2(\operatorname{up}(2x + 1) - \operatorname{up}(2x - 1)). \quad (2.8)$$

Every derivative of F vanishes at 0 and 1, and every derivative of up vanishes at ± 1 .

2.2 Fourier products and moments

We use the angular-frequency convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-i\xi x} dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(\xi) e^{i\xi x} d\xi. \quad (2.9)$$

Then

$$\phi(\xi) := \widehat{\operatorname{up}}(\xi) = \prod_{j=1}^{\infty} \operatorname{sinc}\left(\frac{\xi}{2^j}\right), \quad \operatorname{sinc} z = \frac{\sin z}{z}, \quad (2.10)$$

and

$$\phi(\xi) = \operatorname{sinc}(\xi/2) \phi(\xi/2). \quad (2.11)$$

The one-sided moment generating function is

$$P(s) = \mathbb{E} e^{sY} = \prod_{j=1}^{\infty} \frac{e^{s/2^j} - 1}{s/2^j}, \quad P(2s) = \frac{e^s - 1}{s} P(s). \quad (2.12)$$

Write

$$m_n = \mathbb{E} Y^n, \quad c_k = \mathbb{E} Z^{2k} = \int_{-1}^1 t^{2k} \operatorname{up}(t) dt. \quad (2.13)$$

All these moments are rational. The first centered values are

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \quad c_3 = \frac{583}{59535}. \quad (2.14)$$

The recurrence

$$(2^n - 1)m_n = \sum_{j=0}^{n-1} \binom{n}{j} \frac{m_j}{n - j + 1} \quad (2.15)$$

will be used for numerical coefficients.

2.3 Endpoint rapid variation

The repository's Lambert– W analysis implies, in particular, that F is rapidly varying at 0:

$$\frac{F(tx)}{F(x)} \longrightarrow 0 \quad (x \downarrow 0) \quad \text{for every fixed } 0 \leq t < 1. \quad (2.16)$$

For $x \leq 1/2$, F is convex because differentiating equation (2.7) gives $F''(x) = 4F'(2x) \geq 0$. Consequently

$$F(tx) \leq tF(x), \quad 0 \leq t \leq 1, \quad 0 \leq x \leq \frac{1}{2}. \quad (2.17)$$

The combination of equations (2.16) and (2.17) is the key to the fractional inverse endpoint limits in section 6.

3 Existing ordinary and Volterra calculus: a compact atlas

This section fixes notation and records the strongest formulas already present in the current corpus. They are indispensable inputs, but they are not novelty claims of this report.

3.1 Integer derivative and primitive orbits

Let $\mathcal{F}$ be the signed global extension satisfying $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ and agreeing with F on $(-\infty, 1]$. For $n \in \mathbb{N}_0$,

$$D^n \mathcal{F}(x) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n x). \quad (3.1)$$

For the normalized zero-based Volterra operator

$$(J^n f)(x) = \begin{cases} f(x), & n = 0, \\ \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} f(t) dt, & n \geq 1, \end{cases} \quad (3.2)$$

its exact inverse orbit is

$$J^n \mathcal{F}(x) = 2^{\binom{n}{2}} \mathcal{F}\left(\frac{x}{2^n}\right). \quad (3.3)$$

The same formula holds for bounded F whenever the entire integration segment lies in the agreement range; in particular, for $0 \leq x \leq 1$. Every other n -fold antiderivative differs from equation (3.3) by a polynomial of degree at most $n-1$.

3.2 Monomially weighted primitives

For $x > 0$, $n \geq 1$, and complex a , endpoint flatness makes the following Newton–Volterra series locally normally convergent in (a, x) :

$$\frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} t^a F(t) dt = \sum_{r=0}^{\infty} (-1)^r \binom{n+r-1}{r} a^r x^{a-r} 2^{\binom{n+r}{2}} F\left(\frac{x}{2^{n+r}}\right). \quad (3.4)$$

For $a \in \mathbb{N}_0$ it terminates and is a finite native closed form. Negative and fractional powers are covered by the same convergent series. On $[0, 1]$, weights multiplying $1 - F$ follow by subtracting $J^n(t^a F(t))$ from the elementary Volterra integral of t^a .

3.3 Inverse primitives and ordinary derivatives

For $q = Q(y)$ and $\Re p > 0$,

$$\int_0^y Q(v)^p dv = yq^p - p \int_0^q x^{p-1} F(x) dx, \quad (3.5)$$

$$\int_0^y Q(v) dv = yQ(y) - F\left(\frac{Q(y)}{2}\right). \quad (3.6)$$

The n -fold inverse primitive has the incomplete-moment form

$$I_{0+}^n Q^p(y) = \frac{p}{n!} \int_0^{Q(y)} x^{p-1} (y - F(x))^n dx. \quad (3.7)$$

For ordinary derivatives, introduce

$$L_F = \frac{1}{F'(x)} \frac{d}{dx}. \quad (3.8)$$

Then the exact inverse-chain identity is

$$\frac{d^n}{dy^n} H(Q(y)) = (L_F^n H)(Q(y)). \quad (3.9)$$

In particular, $Q^{(n)} = (L_F^n \text{id}) \circ Q$. Equivalently, differentiating $F(Q(y)) = y$ gives the Bell-polynomial recurrence

$$Q^{(n)}(y) = -\frac{1}{F'(Q(y))} \sum_{k=2}^n F^{(k)}(Q(y)) B_{n,k}(Q'(y), \dots, Q^{(n-k+1)}(y)). \quad (3.10)$$

3.4 Standard Riemann–Liouville primitives

For $\Re \alpha > 0$, using the standard Riemann–Liouville convention [5, 6],

$$(I_{0+}^\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt. \quad (3.11)$$

The corpus already proves the probabilistic formulas

$$I_{0+}^\alpha F(x) = \frac{\mathbb{E}[(x-Y)_+^\alpha]}{\Gamma(\alpha+1)}, \quad (3.12)$$

$$I_{-1+}^\alpha \text{up}(x) = \frac{\mathbb{E}[(x-Z)_+^{\alpha-1}]}{\Gamma(\alpha)}, \quad (3.13)$$

$$I_{0+}^\alpha Q^p(y) = \frac{p}{\Gamma(\alpha+1)} \int_0^{Q(y)} x^{p-1} (y - F(x))^\alpha dx. \quad (3.14)$$

For $\Re \alpha > 1$,

$$I_{0+}^\alpha F(x) = 2^{\alpha-1} I_{0+}^{\alpha-1} F(x/2), \quad (3.15)$$

which interpolates the integer primitive ladder but does not collapse a standard noninteger derivative to a single dilated Fabius value.

4 Exact fractional calculus in the Fabius and inverse clocks

Standard fractional calculus measures memory in the Euclidean difference $x - t$. The Fabius function supplies a second natural clock: probability mass $F(x) - F(t)$. Fractional calculus with respect to a function is classical in general form [7]; the exact Fabius specialization below appears to be absent from the audited corpus.

4.1 Conjugacy definition

Define the composition operator

$$(\mathcal{C}_F h)(y) = h(Q(y)), \quad (\mathcal{C}_F^{-1} g)(x) = g(F(x)). \quad (4.1)$$

Definition 4.1 (Fabius-clock integral and derivative). For $\Re \alpha > 0$,

$$I_F^\alpha = \mathcal{C}_F^{-1} I_{0+}^\alpha \mathcal{C}_F. \quad (4.2)$$

The first-order Fabius-clock derivative is

$$D_F = \mathcal{C}_F^{-1} D \mathcal{C}_F = \frac{1}{F'} \frac{d}{dx}. \quad (4.3)$$

For $n - 1 < \Re \beta < n$, transport the usual Riemann–Liouville and Caputo operators [4, 6] by defining

$${}^{\text{RL}}D_F^\beta = D_F^n I_F^{n-\beta}, \quad (4.4)$$

$${}^{\text{C}}D_F^\beta = I_F^{n-\beta} D_F^n. \quad (4.5)$$

Theorem 4.2 (kernel, semigroup, and exact conjugacy). For suitable h ,

$$(I_F^\alpha h)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (F(x) - F(t))^{\alpha-1} h(t) F'(t) dt. \quad (4.6)$$

Moreover,

$$I_F^\alpha I_F^\gamma = I_F^{\alpha+\gamma}, \quad {}^{\text{C}}D_F^\beta = \mathcal{C}_F^{-1} {}^{\text{C}}D_{0+}^\beta \mathcal{C}_F, \quad {}^{\text{RL}}D_F^\beta = \mathcal{C}_F^{-1} D_{0+}^\beta \mathcal{C}_F. \quad (4.7)$$

Proof. Evaluate equation (4.2) at $y = F(x)$ and substitute $v = F(t)$:

$$\frac{1}{\Gamma(\alpha)} \int_0^{F(x)} (F(x) - v)^{\alpha-1} h(Q(v)) dv = \frac{1}{\Gamma(\alpha)} \int_0^x (F(x) - F(t))^{\alpha-1} h(t) F'(t) dt.$$

Every operator identity is then transported from ordinary fractional calculus by conjugacy. In particular, the semigroup is literal rather than formal. $\square$

4.2 Power laws and Mittag–Leffler eigenfunctions

Theorem 4.3 (diagonal action on powers). For $\Re \lambda > -1$ and $\Re \alpha > 0$,

$$I_F^\alpha F(x)^\lambda = \frac{\Gamma(\lambda + 1)}{\Gamma(\lambda + \alpha + 1)} F(x)^{\lambda+\alpha}. \quad (4.8)$$

For the Riemann–Liouville derivative, wherever both sides are defined,

$${}^{\text{RL}}D_F^\beta F(x)^\lambda = \frac{\Gamma(\lambda + 1)}{\Gamma(\lambda - \beta + 1)} F(x)^{\lambda-\beta}. \quad (4.9)$$

For $n - 1 < \beta < n$, the same formula holds for the Caputo derivative when $\Re \lambda > n - 1$; the usual low-degree polynomial exceptions apply.

Proof. Under $\mathcal{C}_F$, the function $F(x)^\lambda$ becomes y^λ . The assertions are exactly the beta-integral power laws for ordinary Riemann–Liouville and Caputo operators. $\square$

Corollary 4.4 (fractional eigenproblem). For $0 < \beta \leq 1$ and $a \in \mathbb{C}$,

$${}^{\text{C}}D_F^\beta E_\beta(aF(x)^\beta) = a E_\beta(aF(x)^\beta), \quad E_\beta(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\beta k + 1)}. \quad (4.10)$$

Thus the ordinary exponential $e^{aF(x)}$ is recovered at $\beta = 1$.

4.3 The dual inverse clock

The inverse function supplies the dual clock on the probability interval. Define

$$(I_Q^\alpha k)(y) = \frac{1}{\Gamma(\alpha)} \int_0^y (Q(y) - Q(v))^{\alpha-1} k(v) Q'(v) dv, \quad D_Q = \frac{1}{Q'} \frac{d}{dy}. \quad (4.11)$$

This is conjugate to ordinary fractional calculus by $x = Q(y)$. Hence

$$I_Q^\alpha Q(y)^\lambda = \frac{\Gamma(\lambda + 1)}{\Gamma(\lambda + \alpha + 1)} Q(y)^{\lambda + \alpha}, \quad (4.12)$$

$${}^{\text{RL}}D_Q^\beta Q(y)^\lambda = \frac{\Gamma(\lambda + 1)}{\Gamma(\lambda - \beta + 1)} Q(y)^{\lambda - \beta}. \quad (4.13)$$

The pair (I_F, D_F) and (I_Q, D_Q) is therefore exactly dual under the mutually inverse coordinate maps F and Q .

Remark 4.5 (What this does and does not say). The Fabius-clock derivative is not the standard fractional derivative of F as a function of Euclidean x . It is a nonlinear-coordinate operator in which the intricate Fabius geometry is moved into the kernel. Its advantage is exact algebra; the standard Caputo and Weyl derivatives studied below retain the original Euclidean clock and are analytically more delicate.

5 Caputo and Riemann–Liouville transmutation through the inverse

The ordinary inverse-chain rule [equation \(3.9\)](#) has a nonlocal fractional counterpart. It converts a singular derivative of $H \circ Q$ in the y variable into an integral over the much better behaved x variable.

5.1 Left fractional integrals and derivatives

Theorem 5.1 (general inverse-composition transmutation). *Let $q = Q(y)$ and $\Re \alpha > 0$. If H is integrable against the displayed kernels, then*

$$I_{0+}^\alpha [H \circ Q](y) = \frac{1}{\Gamma(\alpha)} \int_0^q (y - F(x))^{\alpha-1} H(x) F'(x) dx, \quad (5.1)$$

$$= \frac{H(0)y^\alpha}{\Gamma(\alpha + 1)} + \frac{1}{\Gamma(\alpha + 1)} \int_0^q (y - F(x))^\alpha H'(x) dx, \quad (5.2)$$

where the second identity assumes absolute continuity of H . For $0 < \beta < 1$,

$${}^{\text{C}}D_{0+}^\beta [H \circ Q](y) = \frac{1}{\Gamma(1 - \beta)} \int_0^q (y - F(x))^{-\beta} H'(x) dx, \quad (5.3)$$

and

$$D_{0+}^\beta [H \circ Q](y) = \frac{H(0)y^{-\beta}}{\Gamma(1 - \beta)} + {}^{\text{C}}D_{0+}^\beta [H \circ Q](y). \quad (5.4)$$

Proof. Substitute $v = F(x)$ in the definition of $I_{0+}^\alpha [H(Q(v))]$ to obtain [equation \(5.1\)](#). Since

$$d(y - F(x))^\alpha = -\alpha(y - F(x))^{\alpha-1} F'(x) dx,$$

integration by parts gives [equation \(5.2\)](#). For the Caputo derivative,

$$\frac{d}{dv} H(Q(v)) = H'(Q(v)) Q'(v).$$

After $v = F(x)$, the factors $Q'(F(x))F'(x)$ cancel exactly, leaving [equation \(5.3\)](#). The final identity is the standard Riemann–Liouville/Caputo endpoint correction. $\square$

Corollary 5.2 (powers of the inverse Fabius function). *For $r > 0$ and $0 < \beta < 1$,*

$${}^C D_{0+}^\beta Q(y)^r = \frac{r}{\Gamma(1-\beta)} \int_0^{Q(y)} x^{r-1} (y - F(x))^{-\beta} dx. \quad (5.5)$$

Because $Q(0) = 0$, the Riemann–Liouville and Caputo derivatives coincide for Q^r in this range whenever the integral is finite.

5.2 All positive noninteger orders

Let $n - 1 < \beta < n$. From equation (3.9),

$$\frac{d^n}{dy^n} [H(Q(y))] = (L_F^n H)(Q(y)). \quad (5.6)$$

Therefore:

Proposition 5.3 (higher-order Caputo transmutation). *Suppose $H \circ Q$ belongs to the classical Caputo domain on $[0, y]$ and $(L_F^n H)(x)F'(x)$ is integrable against the displayed kernel. Then*

$${}^C D_{0+}^\beta [H \circ Q](y) = \frac{1}{\Gamma(n-\beta)} \int_0^{Q(y)} (y - F(x))^{n-\beta-1} (L_F^n H)(x) F'(x) dx. \quad (5.7)$$

The Riemann–Liouville derivative is obtained by adding the usual finite endpoint jet

$$\sum_{k=0}^{n-1} \frac{(H \circ Q)^{(k)}(0+)}{\Gamma(k-\beta+1)} y^{k-\beta}, \quad (5.8)$$

when these endpoint values exist.

Higher-order inverse derivatives

The inverse Fabius function has singular derivatives at 0 and 1. Consequently, a formula obtained by formally inserting $H(x) = x$ into equation (5.7) need not define a classical Caputo derivative for every $\beta > 1$. The theorem is deliberately stated in terms of the actual Caputo domain. Distributional or weighted-space extensions are promising, but require a separate endpoint-domain analysis.

5.3 Right-sided transmutation

For $0 < \beta < 1$ and $q = Q(y)$, the right-sided counterparts are

$$I_{1-}^\alpha [H \circ Q](y) = \frac{H(1)(1-y)^\alpha}{\Gamma(\alpha+1)} - \frac{1}{\Gamma(\alpha+1)} \int_q^1 (F(x) - y)^\alpha H'(x) dx, \quad (5.9)$$

$${}^C D_{1-}^\beta [H \circ Q](y) = -\frac{1}{\Gamma(1-\beta)} \int_q^1 (F(x) - y)^{-\beta} H'(x) dx, \quad (5.10)$$

$$D_{1-}^\beta [H \circ Q](y) = \frac{H(1)(1-y)^{-\beta}}{\Gamma(1-\beta)} + {}^C D_{1-}^\beta [H \circ Q](y). \quad (5.11)$$

They follow by the same substitution and integration by parts. Symmetry $Q(1-y) = 1 - Q(y)$ converts left endpoint statements into right endpoint statements without recomputing the Fabius asymptotics.

6 Fractional endpoint laws for inverse powers

The integral formulas above permit endpoint asymptotics without differentiating the Lambert– W inverse expansion term by term.

Theorem 6.1 (fractional primitive and Caputo endpoint laws). *Let $r > 0$ and $\alpha > 0$. Then*

$$I_{0+}^{\alpha} Q(y)^r \sim \frac{Q(y)^r y^{\alpha}}{\Gamma(\alpha + 1)} \quad (y \downarrow 0). \quad (6.1)$$

For $0 < \beta < 1$,

$${}^C D_{0+}^{\beta} Q(y)^r \sim \frac{Q(y)^r y^{-\beta}}{\Gamma(1 - \beta)} \quad (y \downarrow 0). \quad (6.2)$$

Proof. Put $q = Q(y)$ and scale $x = qt$ in equation (3.14). The ratio of the two sides of equation (6.1) is

$$r \int_0^1 t^{r-1} \left(1 - \frac{F(qt)}{F(q)} \right)^{\alpha} dt. \quad (6.3)$$

For each fixed $t < 1$, rapid variation equation (2.16) makes the parenthesis tend to 1. The integrand is bounded by rt^{r-1} , so dominated convergence gives 1.

Likewise, equation (5.5) yields the ratio

$$r \int_0^1 t^{r-1} \left(1 - \frac{F(qt)}{F(q)} \right)^{-\beta} dt. \quad (6.4)$$

For $q \leq 1/2$, convexity gives $F(qt)/F(q) \leq t$; hence the integrand is bounded by $rt^{r-1}(1-t)^{-\beta}$, which is integrable because $r > 0$ and $\beta < 1$. Dominated convergence again gives 1. $\square$

Corollary 6.2 (right endpoint). *As $y \uparrow 1$, the corresponding right-sided laws hold for $(1 - Q(y))^r$ with $1 - y$ in place of y .*

Remark 6.3 (Slow convergence). The limit in equation (6.2) is rigorous but numerically slow. For $r = 1$, $\beta = 0.4$, the ratio in equation (6.4) decreases only from approximately 1.1835 at $y = 10^{-2}$ to 1.1127 at $y = 10^{-7}$. This is consistent with a first correction on the Lambert scale $\rho(y)^{-1}$, where

$$\rho(y) = \sqrt{\frac{2 \log(1/y)}{\log 2}}. \quad (6.5)$$

A phase-resolved conjecture is stated in theorem 16.1.

7 A standard fractional bridge from Fabius to Rvachev

Because F is flat at 0, its left Riemann–Liouville and Caputo derivatives coincide for every positive noninteger order. Unlike the Fabius-clock operator, the following theorem concerns the ordinary Euclidean clock.

Theorem 7.1 (all-order fractional F -up bridge). *Let $n - 1 < \beta < n$ and $0 < x < 1$. Then*

$$D_{0+}^{\beta} F(x) = {}^C D_{0+}^{\beta} F(x) = 2^{\beta} \left(I_{-1+}^{n-\beta} \text{up}^{(n-1)} \right) (2x - 1). \quad (7.1)$$

In particular, for $0 < \beta < 1$,

$$D_{0+}^{\beta} F(x) = 2^{\beta} \left(I_{-1+}^{1-\beta} \text{up} \right) (2x - 1). \quad (7.2)$$

Proof. Flatness removes every endpoint correction between the two fractional derivatives. Since $F'(t) = 2 \operatorname{up}(2t - 1)$,

$$F^{(n)}(t) = 2^n \operatorname{up}^{(n-1)}(2t - 1). \quad (7.3)$$

Insert this into $I_{0+}^{n-\beta} F^{(n)}(x)$ and substitute $s = 2t - 1$. The factors are 2^n from differentiation, 2^{-1} from dt , and $2^{-(n-\beta-1)}$ from the kernel, leaving exactly 2^β . $\square$

Remark 7.2 (Interpolation endpoints). For $0 < \beta < 1$, letting $\beta \downarrow 0$ in equation (7.2) gives

$$F(x) = \int_{-1}^{2x-1} \operatorname{up}(t) dt,$$

whereas $\beta \uparrow 1$ gives $F'(x) = 2 \operatorname{up}(2x - 1)$. Thus the identity is a literal continuum between the CDF and its density.

7.1 Fractional derivatives of up from the support boundary

On the left half of its support, $\operatorname{up}(x) = F(x + 1)$. Therefore, for $-1 < x < 0$,

$$D_{-1+}^\beta \operatorname{up}(x) = 2^\beta \left(I_{-1+}^{n-\beta} \operatorname{up}^{(n-1)} \right) (2x + 1), \quad n - 1 < \beta < n. \quad (7.4)$$

On the right half, reflection gives

$$D_{1-}^\beta \operatorname{up}(x) = 2^\beta \left(I_{-1+}^{n-\beta} \operatorname{up}^{(n-1)} \right) (1 - 2x), \quad 0 < x < 1, \quad (7.5)$$

with the standard right-Caputo sign convention. These are boundary-anchored identities; they should not be confused with the translation-invariant Weyl operators in the next section.

7.2 Fractional primitives of $1 - F$

On $[0, 1]$, $\operatorname{up}(x) = 1 - F(x)$. Hence

$$I_{0+}^\alpha \operatorname{up}(x) = \frac{x^\alpha}{\Gamma(\alpha + 1)} - I_{0+}^\alpha F(x), \quad (7.6)$$

and the probabilistic formula equation (3.12) gives

$$I_{0+}^\alpha \operatorname{up}(x) = \frac{x^\alpha - \mathbb{E}[(x - Y)_+^\alpha]}{\Gamma(\alpha + 1)}. \quad (7.7)$$

At $x = 1$, symmetry yields

$$I_{0+}^\alpha F(1) = \frac{\mathcal{M}(\alpha)}{\Gamma(\alpha + 1)}, \quad \mathcal{M}(s) = \mathbb{E}Y^s = \int_0^1 Q(y)^s dy, \quad (7.8)$$

so endpoint fractional integration is exactly the entire inverse-moment function already developed in the corpus.

8 Whole-line Weyl derivatives and complex-order refinement

Boundary-anchored fractional derivatives break translation invariance. The compact support and entire Fourier product of up make Weyl calculus the natural whole-line alternative.

8.1 Fourier-multiplier definition

Following the standard Weyl multiplier conventions [5], choose principal branches and define, initially on Schwartz functions and then by tempered-distribution continuation,

$$\mathfrak{D}_+^\sigma f = \mathcal{F}^{-1}[(i\xi)^\sigma \widehat{f}(\xi)], \quad (8.1)$$

$$\mathfrak{D}_-^\sigma f = \mathcal{F}^{-1}[(-i\xi)^\sigma \widehat{f}(\xi)]. \quad (8.2)$$

For $\Re\alpha > 0$, the left Weyl integral is

$$(\mathfrak{I}_+^\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_{-\infty}^x (x-t)^{\alpha-1} f(t) dt, \quad (8.3)$$

whose multiplier is $(i\xi)^{-\alpha}$ in the boundary-value sense.

8.2 The complex-order hierarchy

Theorem 8.1 (Weyl–Rvachev refinement hierarchy). *Let*

$$W_\sigma = \mathfrak{D}_+^\sigma \text{ up}. \quad (8.4)$$

For every complex σ for which the terms are interpreted consistently as tempered distributions,

$$W_\sigma(x) = 2^\sigma (W_{\sigma-1}(2x+1) - W_{\sigma-1}(2x-1)). \quad (8.5)$$

If $n-1 < \beta < n$, this becomes

$$\mathfrak{D}_+^\beta \text{ up}(x) = 2^\beta \left[(\mathfrak{I}_+^{n-\beta} \text{ up}^{(n-1)})(2x+1) - (\mathfrak{I}_+^{n-\beta} \text{ up}^{(n-1)})(2x-1) \right]. \quad (8.6)$$

At $\beta = 1$ it reduces to Rvachev’s differential–difference formula (2.8).

Proof. Apply $\mathfrak{D}_+^{\sigma-1}$ to equation (2.8). Positive dilation obeys

$$\mathfrak{D}_+^\alpha [f(2x+a)] = 2^\alpha (\mathfrak{D}_+^\alpha f)(2x+a).$$

The prefactor 2 in equation (2.8) therefore produces 2^σ . Equivalently, multiply the Fourier refinement equation (2.11) by $(i\xi)^\sigma$ and invert. Finally, $W_{\beta-1} = \mathfrak{I}_+^{n-\beta} \text{ up}^{(n-1)}$ by equality of Fourier multipliers. $\square$

Corollary 8.2 (right hierarchy and reflection). *Because up is even,*

$$\mathfrak{D}_-^\sigma \text{ up}(x) = \mathfrak{D}_+^\sigma \text{ up}(-x). \quad (8.7)$$

The corresponding right-sided refinement is obtained by reflecting equation (8.5).

8.3 Riesz symmetrization

For real $0 < \beta < 2$, define the Riesz fractional Laplacian independently by

$$(-\Delta)^{\beta/2} f(\xi) = |\xi|^\beta \widehat{f}(\xi). \quad (8.8)$$

When $\cos(\pi\beta/2) \neq 0$,

$$(-\Delta)^{\beta/2} f = \frac{\mathfrak{D}_+^\beta f + \mathfrak{D}_-^\beta f}{2 \cos(\pi\beta/2)}. \quad (8.9)$$

At $\beta = 1$ the quotient is understood by the Fourier definition (or via the Hilbert-transform derivative), not by separately evaluating the vanishing numerator and denominator.

9 Generalized Stieltjes transforms of complex order

The ordinary Cauchy transform is only the order-1 member of a much richer family. For real $z > 1$ and $\lambda \in \mathbb{C}$, define

$$S_\lambda(z) = \int_{-1}^1 \frac{\text{up}(t)}{(z-t)^\lambda} dt = \mathbb{E}(z-Z)^{-\lambda}, \quad (9.1)$$

where the real logarithm fixes the power. Analytic continuation in z is available on the slit plane after choosing a branch.

9.1 Moment, derivative, and dyadic-order hierarchies

Theorem 9.1 (complete generalized Stieltjes hierarchy). *For fixed $z > 1$, $S_\lambda(z)$ is entire in λ . For $|z| > 1$,*

$$S_\lambda(z) = z^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda)_{2k}}{(2k)!} \frac{c_k}{z^{2k}}, \quad (9.2)$$

locally normally in (λ, z) . Its z derivatives satisfy

$$\frac{d^n}{dz^n} S_\lambda(z) = (-1)^n (\lambda)_n S_{\lambda+n}(z). \quad (9.3)$$

Moreover,

$$S_\lambda(z) = 2^{\lambda-1} \int_{-1}^1 S_\lambda(2z-v) dv, \quad (9.4)$$

and, for $\lambda \neq 1$,

$$S_\lambda(z) = \frac{2^{\lambda-1}}{1-\lambda} (S_{\lambda-1}(2z+1) - S_{\lambda-1}(2z-1)). \quad (9.5)$$

At $\lambda = 1$, if

$$L(z) = \mathbb{E} \log(z-Z), \quad (9.6)$$

then

$$S_1(z) = L(2z+1) - L(2z-1). \quad (9.7)$$

Proof. For $z > 1$, $\log(z-Z)$ is bounded, so $S_\lambda(z) = \mathbb{E} \exp[-\lambda \log(z-Z)]$ is entire. Expanding $(1-Z/z)^{-\lambda}$ and using symmetry removes odd moments, proving [equation \(9.2\)](#). Differentiation under the integral gives [equation \(9.3\)](#).

The random series satisfies

$$Z \stackrel{d}{=} \frac{V + Z'}{2}, \quad V \sim \text{Unif}[-1, 1], \quad (9.8)$$

with Z' an independent copy. Conditioning on V gives [equation \(9.4\)](#). Changing variables turns its integral into $\int_{2z-1}^{2z+1} S_\lambda(w) dw$. Since $S'_{\lambda-1} = (1-\lambda)S_\lambda$, this proves [equation \(9.5\)](#). Taking the removable limit at $\lambda = 1$ and using $L' = S_1$ gives [equation \(9.7\)](#). $\square$

Remark 9.2 (Interpolation between Appell polynomials and resolvents). For $m \in \mathbb{N}_0$,

$$S_{-m}(z) = \mathbb{E}(z-Z)^m \quad (9.9)$$

is a rational-coefficient Appell polynomial. Positive integer orders are ordinary higher resolvents. Thus [equation \(9.5\)](#) is a single complex-order hierarchy connecting the polynomial and resolvent sides.

9.2 One-sided version

Define, for $z > 1$,

$$R_\lambda(z) = \mathbb{E}(z - Y)^{-\lambda}. \quad (9.10)$$

Since $Y \stackrel{d}{=} (U + Y')/2$ with U uniform on $[0, 1]$,

$$R_\lambda(z) = 2^\lambda \int_0^1 R_\lambda(2z - u) \, du, \quad (9.11)$$

$$= \frac{2^\lambda}{1 - \lambda} (R_{\lambda-1}(2z) - R_{\lambda-1}(2z - 1)), \quad \lambda \neq 1. \quad (9.12)$$

The centered and one-sided transforms are related by

$$S_\lambda(z) = 2^{-\lambda} R_\lambda\left(\frac{z+1}{2}\right). \quad (9.13)$$

For $\lambda > 0$, S_λ and R_λ are completely monotone on their real domains; all alternating derivatives are explicitly positive by [equation \(9.3\)](#).

10 Complex-order resolvents involving F and the inverse

10.1 Distribution and survival resolvents

For $z > 1$ and $\lambda \neq 1$, Fubini or integration by parts gives

$$\int_0^1 \frac{F(x)}{(z-x)^\lambda} \, dx = \frac{R_{\lambda-1}(z) - (z-1)^{1-\lambda}}{1-\lambda}, \quad (10.1)$$

$$\int_0^1 \frac{1-F(x)}{(z-x)^\lambda} \, dx = \frac{z^{1-\lambda} - R_{\lambda-1}(z)}{1-\lambda}. \quad (10.2)$$

At $\lambda = 1$, the removable limits are

$$\int_0^1 \frac{F(x)}{z-x} \, dx = \mathbb{E} \log(z - Y) - \log(z - 1), \quad (10.3)$$

$$\int_0^1 \frac{1-F(x)}{z-x} \, dx = \log z - \mathbb{E} \log(z - Y). \quad (10.4)$$

These extend the integer resolvent formulas in the corpus to arbitrary complex order.

10.2 Inverse-composition resolvents

Theorem 10.1 (complex inverse-resolvent transmutation). *Let H be absolutely continuous on $[0, 1]$, $z > 1$, and $\lambda \neq 1$. Then*

$$T_{\lambda,H}(z) := \int_0^1 (z-y)^{-\lambda} H(Q(y)) \, dy \quad (10.5)$$

$$= \frac{H(0)z^{1-\lambda} - H(1)(z-1)^{1-\lambda}}{1-\lambda} + \frac{1}{1-\lambda} \int_0^1 (z-F(x))^{1-\lambda} H'(x) \, dx. \quad (10.6)$$

At $\lambda = 1$,

$$T_{1,H}(z) = H(0) \log z - H(1) \log(z-1) + \int_0^1 H'(x) \log(z-F(x)) \, dx. \quad (10.7)$$

Furthermore,

$$\frac{d^n}{dz^n} T_{\lambda,H}(z) = (-1)^n (\lambda)_n T_{\lambda+n,H}(z). \quad (10.8)$$

Proof. Substitute $y = F(x)$:

$$T_{\lambda,H}(z) = \int_0^1 (z - F(x))^{-\lambda} H(x) F'(x) dx.$$

Since $d(z - F(x))^{1-\lambda} = -(1-\lambda)(z - F(x))^{-\lambda} F'(x) dx$, integration by parts proves equation (10.6). Expanding at $1 - \lambda = 0$ gives equation (10.7); differentiation under the integral gives the hierarchy. $\square$

For $H(x) = x^p$ with $p > 0$,

$$\int_0^1 (z - y)^{-\lambda} Q(y)^p dy = \frac{-(z - 1)^{1-\lambda} + p \int_0^1 x^{p-1} (z - F(x))^{1-\lambda} dx}{1 - \lambda}. \quad (10.9)$$

This is the complex-order extension of the inverse logarithmic resolvent found in the latest corpus report.

10.3 Large- z mixed-moment series

For $|z| > 1$,

$$T_{\lambda,H}(z) = z^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda)_n}{n! z^n} \int_0^1 y^n H(Q(y)) dy. \quad (10.10)$$

For $H(x) = x^p$, integration by parts after $y = F(x)$ gives

$$\int_0^1 y^n Q(y)^p dy = \frac{1}{n+1} - \frac{p}{n+1} \int_0^1 x^{p-1} F(x)^{n+1} dx. \quad (10.11)$$

Thus every coefficient is reduced to a beta-type Fabius integral. When p and n are nonnegative integers, finite dyadic formulas and rational moments can be inserted wherever the existing corpus machinery applies.

11 Riesz potentials and the fractional Laplacian of up

This section gives the cleanest new transform consequence. Compact support turns the nonlocal exterior fractional Laplacian into the generalized Stieltjes hierarchy exactly, not merely asymptotically.

11.1 Exact exterior fractional Laplacian

For $0 < \beta < 2$, use the standard one-dimensional normalization [8]:

$$C_{1,\beta} = \frac{\beta 2^{\beta-1} \Gamma((1+\beta)/2)}{\sqrt{\pi} \Gamma(1-\beta/2)}. \quad (11.1)$$

Then

$$(-\Delta)^{\beta/2} f(x) = C_{1,\beta} \text{PV} \int_{\mathbb{R}} \frac{f(x) - f(t)}{|x - t|^{1+\beta}} dt. \quad (11.2)$$

Theorem 11.1 (exact exterior Stieltjes representation). *For $0 < \beta < 2$ and $|x| > 1$,*

$$(-\Delta)^{\beta/2} \text{up}(x) = -C_{1,\beta} S_{1+\beta}(|x|). \quad (11.3)$$

Consequently,

$$(-\Delta)^{\beta/2} \text{up}(x) = -C_{1,\beta} |x|^{-1-\beta} \sum_{k=0}^{\infty} \frac{(1+\beta)_{2k}}{(2k)!} \frac{c_k}{|x|^{2k}}. \quad (11.4)$$

The first terms are

$$(-\Delta)^{\beta/2} \text{up}(x) = -C_{1,\beta} |x|^{-1-\beta} \left[1 + \frac{(1+\beta)(2+\beta)}{18x^2} + \frac{19(1+\beta)(2+\beta)(3+\beta)(4+\beta)}{16200x^4} + O(x^{-6}) \right]. \quad (11.5)$$

Proof. For $|x| > 1$, $\text{up}(x) = 0$ and the singularity is absent. Thus

$$(-\Delta)^{\beta/2} \text{up}(x) = -C_{1,\beta} \int_{-1}^1 \frac{\text{up}(t)}{|x-t|^{1+\beta}} dt.$$

For $x > 1$ this is $-C_{1,\beta} S_{1+\beta}(x)$. Evenness gives the same formula with $|x|$ for $x < -1$. Insert [equation \(9.2\)](#) and the exact moments [equation \(2.14\)](#). $\square$

Corollary 11.2 (sign and conservation). *The exterior fractional Laplacian is strictly negative. At the center,*

$$\begin{aligned} (-\Delta)^{\beta/2} \text{up}(0) &= C_{1,\beta} \int_{\mathbb{R}} \frac{1 - \text{up}(t)}{|t|^{1+\beta}} dt \\ &= 2C_{1,\beta} \left(\int_0^1 \frac{1 - \text{up}(t)}{t^{1+\beta}} dt + \frac{1}{\beta} \right) > 0. \end{aligned} \quad (11.6)$$

Moreover,

$$\int_{\mathbb{R}} (-\Delta)^{\beta/2} \text{up}(x) dx = 0. \quad (11.7)$$

Hence at least one zero occurs on each half-axis.

Proof. The exterior sign follows from [equation \(11.3\)](#). Since $\text{up}(0) = 1$, the singular-integral definition at 0 gives the first line of [equation \(11.6\)](#); splitting at $|t| = 1$ gives the second. The function $(-\Delta)^{\beta/2} \text{up}$ is integrable because it is smooth on compact sets and has the exterior decay [equation \(11.4\)](#). Its integral is therefore its Fourier transform at 0, namely $|0|^\beta \widehat{\text{up}}(0) = 0$. Evenness and the opposite signs at 0 and outside the support force a zero on each half-axis. $\square$

11.2 Riesz potentials

For $0 < \alpha < 1$, the Riesz potential with multiplier $|\xi|^{-\alpha}$ is

$$(I_\alpha^R f)(x) = \kappa_\alpha \int_{\mathbb{R}} |x-t|^{\alpha-1} f(t) dt, \quad \kappa_\alpha = \frac{\Gamma((1-\alpha)/2)}{2^\alpha \sqrt{\pi} \Gamma(\alpha/2)}. \quad (11.8)$$

Theorem 11.3 (exterior Riesz-potential expansion). *For $0 < \alpha < 1$ and $|x| > 1$,*

$$I_\alpha^R \text{up}(x) = \kappa_\alpha S_{1-\alpha}(|x|) = \kappa_\alpha |x|^{\alpha-1} \sum_{k=0}^{\infty} \frac{(1-\alpha)_{2k}}{(2k)!} \frac{c_k}{|x|^{2k}}. \quad (11.9)$$

In particular, the leading far-field coefficient is exactly the unit mass of up ; all corrections are rational polynomials in α .

11.3 A numerical one-node phenomenon

The accompanying experiment reconstructs up from the infinite sinc product, then evaluates the nonperiodic singular integral [equation \(11.2\)](#). A symmetric second-difference removes the principal-value singularity and a Gauss–Jacobi rule integrates the remaining $u^{1-\beta}$ weight. On a 601-point profile grid, exactly one positive sign change in $(0, 1)$ is resolved for each tested order; Brent refinement gives the following approximations (the displayed seven decimal places are stable under a moderate change of quadrature order):

β	positive zero a_β
0.25	0.6721595
0.50	0.6100983
0.75	0.5741282
1.00	0.5506118
1.25	0.5344913
1.50	0.5232645
1.75	0.5154602

Conjecture 11.4 (one-node law). *For every $0 < \beta < 2$, the even function $(-\Delta)^{\beta/2} \text{up}$ has exactly two simple zeros $\pm a_\beta$. It is positive on $(-a_\beta, a_\beta)$ and negative outside. Moreover, a_β is strictly decreasing, with*

$$\lim_{\beta \downarrow 0} a_\beta = 1, \quad \lim_{\beta \uparrow 2} a_\beta = \frac{1}{2}. \quad (11.10)$$

The endpoint $1/2$ is natural: on $(0, 1/2)$, $-\text{up}'' = F'' = 4F'(2x) > 0$, while the integer second derivative changes its Thue–Morse cell sign across $x = 1/2$. Proving uniqueness for noninteger order will likely require a variation-diminishing or total-positivity principle for the singular kernel combined with the special unimodality of up .

12 Fractional energies and autocorrelation transforms

The integer derivative norm law in the corpus samples a continuous spectral Mellin transform at even integers. Define

$$\mathcal{E}(s) = \frac{1}{2\pi} \int_{\mathbb{R}} |\xi|^s |\phi(\xi)|^2 d\xi, \quad \Re s > -1. \quad (12.1)$$

The sinc product decays faster than every power, so this is holomorphic in the half-plane $\Re s > -1$.

Proposition 12.1 (log-convex fractional energy). *For real $s > -1$, $\mathcal{E}(s) > 0$ and $\log \mathcal{E}(s)$ is convex. At even integers,*

$$\mathcal{E}(2n) = \left\| \text{up}^{(n)} \right\|_2^2 = 2^{n(n+1)} \mathcal{E}(0). \quad (12.2)$$

Consequently, for $2n \leq s \leq 2n + 2$,

$$\mathcal{E}(s) \leq \mathcal{E}(0) 2^{(n+1)(s-n)}. \quad (12.3)$$

Proof. Positivity is immediate. Hölder’s inequality applied to $|\xi|^s |\phi|^2 d\xi$ proves log-convexity. Parseval gives the first equality in [equation \(12.2\)](#); the second is the existing derivative-equimeasurability law. Convex interpolation between the values at $2n$ and $2n + 2$ gives [equation \(12.3\)](#). $\square$

For a shift $a \in \mathbb{R}$, define the fractional autocorrelation

$$\mathcal{C}_s(a) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\xi a} |\xi|^s |\phi(\xi)|^2 d\xi. \quad (12.4)$$

Since up is even,

$$\mathcal{C}_s(a) = (-\Delta_a)^{s/2} (\text{up} * \text{up})(a), \quad \mathcal{C}_s(0) = \mathcal{E}(s). \quad (12.5)$$

Thus the entire family of fractional Sobolev energies and shifted correlations is encoded by the same lacunary sinc product. For $s = 2n$, this reduces to integer derivatives; for noninteger s , it is a new continuous interpolation problem closely related to the shifted Stieltjes–Wigert spectral measure in the derivative-norm report.

13 Definite integrals and transform catalog

This section gathers a practical set of exact reductions. Most of the first group is already in the corpus; several composition identities at the end are new elementary consequences.

13.1 CDF–survival master identities

Let G be absolutely continuous on $[0, 1]$. Then

$$\int_0^1 G'(x)F(x) \, dx = G(1) - \mathbb{E}G(Y), \quad (13.1)$$

$$\int_0^1 G'(x)(1 - F(x)) \, dx = \mathbb{E}G(Y) - G(0). \quad (13.2)$$

Examples include

$$\int_0^1 x^{s-1}F(x) \, dx = \frac{1 - \mathcal{M}(s)}{s}, \quad (13.3)$$

$$\int_0^1 x^{s-1} \text{up}(x) \, dx = \frac{\mathcal{M}(s)}{s}, \quad (13.4)$$

$$\int_0^1 e^{tx}F(x) \, dx = \frac{e^t - P(t)}{t}, \quad (13.5)$$

$$\int_0^1 e^{tx} \text{up}(x) \, dx = \frac{P(t) - 1}{t}. \quad (13.6)$$

The apparent singularities at $s = 0$ or $t = 0$ are removable. Differentiating in the parameters generates logarithmic and polynomially weighted families.

13.2 Quantile transport and power duality

For every integrable H ,

$$\int_0^1 H(Q(y)) \, dy = \int_0^1 H(x)F'(x) \, dx = \mathbb{E}H(Y). \quad (13.7)$$

For $\Re p, \Re q > 0$,

$$p \int_0^1 x^{p-1}F(x)^q \, dx + q \int_0^1 y^{q-1}Q(y)^p \, dy = 1. \quad (13.8)$$

The universal beta family is

$$\int_0^1 F(x)^a(1 - F(x))^b F'(x) \, dx = B(a + 1, b + 1), \quad \Re a, \Re b > -1. \quad (13.9)$$

Using $F' = 2 \text{up}(2x - 1)$,

$$\int_0^1 F(x)^a \text{up}(x)^b \text{up}(2x - 1) \, dx = \frac{1}{2}B(a + 1, b + 1). \quad (13.10)$$

This gives, for example,

$$\int_0^1 F(x)(1 - F(x))F'(x) \, dx = \frac{1}{6}. \quad (13.11)$$

13.3 Full-line analytic kernels and Fourier transforms

If $h(t) = \sum_{n \geq 0} a_n t^n$ is entire, then

$$\int_{-1}^1 h(t) \operatorname{up}(t) dt = \sum_{k=0}^{\infty} a_{2k} c_k. \quad (13.12)$$

The Fourier, bilateral Laplace, Bessel, and generalized Stieltjes transforms are therefore

$$\int_{-1}^1 e^{i\xi t} \operatorname{up}(t) dt = \prod_{j \geq 1} \operatorname{sinc}(\xi/2^j), \quad (13.13)$$

$$\int_{-1}^1 e^{st} \operatorname{up}(t) dt = \prod_{j \geq 1} \frac{\sinh(s/2^j)}{s/2^j}, \quad (13.14)$$

$$\int_{-1}^1 J_0(at) \operatorname{up}(t) dt = \sum_{k=0}^{\infty} \frac{(-1)^k (a^2/4)^k}{(k!)^2} c_k, \quad (13.15)$$

$$\int_{-1}^1 (z-t)^{-\lambda} \operatorname{up}(t) dt = S_{\lambda}(z). \quad (13.16)$$

13.4 Product and order-statistic integrals

Let $m, n \geq 1$, let $Y_1, Y_2, \dots$ be independent copies of Y , $A_m = \max(Y_1, \dots, Y_m)$, and $B_n = \min(Y_{m+1}, \dots, Y_{m+n})$. Then

$$F(x)^m (1 - F(x))^n = \mathbb{P}(A_m \leq x < B_n), \quad (13.17)$$

and hence, for real $p > -1$,

$$\int_0^1 x^p F(x)^m (1 - F(x))^n dx = \frac{1}{p+1} \mathbb{E} [(B_n^{p+1} - A_m^{p+1})_+]. \quad (13.18)$$

This is an exact probabilistic closed form even when no reduction to rational moments is available. The Lambert endpoint theory controls its large- m or large- n regimes.

13.5 Compositions with the inverse density

Equation 2.6 yields a useful family that appears not to have been isolated in the audited integral reports. For $p > -1$,

$$\int_0^1 (F'(Q(y)))^p dy = \int_0^1 F'(x)^{p+1} dx = 2^p \int_{-1}^1 \operatorname{up}(t)^{p+1} dt, \quad (13.19)$$

$$\int_0^1 \operatorname{up}(2Q(y) - 1)^p dy = \int_{-1}^1 \operatorname{up}(t)^{p+1} dt, \quad (13.20)$$

$$\int_0^1 Q'(y)^{-p} dy = 2^p \|\operatorname{up}\|_{p+1}^{p+1}. \quad (13.21)$$

The proof is the single substitution $y = F(x)$ followed by $t = 2x - 1$. These formulas tie inverse-derivative integrals directly to the complete L^p spectrum of the compact density.

14 Indefinite integral catalog

To avoid ambiguity, an “indefinite integral” below means the normalized primitive with zero initial data; arbitrary constants or lower-degree polynomials can be restored afterward.

14.1 Fabius and reflected Fabius functions

For $0 \leq x \leq 1$,

$$\int_0^x F(t) dt = F(x/2), \quad (14.1)$$

$$J^n F(x) = 2^{\binom{n}{2}} F(x/2^n), \quad (14.2)$$

$$\int_0^x (1 - F(t)) dt = x - F(x/2), \quad (14.3)$$

$$J^n(1 - F)(x) = \frac{x^n}{n!} - 2^{\binom{n}{2}} F(x/2^n). \quad (14.4)$$

Because $F(1 - x) = 1 - F(x)$, these are also primitives of the reflected Fabius function. Polynomial and arbitrary complex-power weights are supplied by [equation \(3.4\)](#).

14.2 Rvachev folds

On $[-1, 0]$, $\text{up}(x) = F(x + 1)$, while on $[0, 1]$, $\text{up}(x) = 1 - F(x)$. Thus a normalized first primitive based at -1 is

$$\int_{-1}^x \text{up}(t) dt = \begin{cases} F((x + 1)/2), & -1 \leq x \leq 0, \\ 1 - F((1 - x)/2), & 0 \leq x \leq 1, \\ 1, & x \geq 1. \end{cases} \quad (14.5)$$

Higher integer primitives follow either by repeated use of [equation \(14.2\)](#) on each fold or from the exact truncated moment

$$I_{-1+}^n \text{up}(x) = \frac{1}{(n - 1)!} \mathbb{E}[(x - Z)_+^{n-1}]. \quad (14.6)$$

For $x \geq 1$ this becomes an Appell polynomial in x with rational coefficients.

14.3 Inverse Fabius function

The normalized first primitive is [equation \(3.6\)](#). More generally, for $\alpha > 0$ and $p > 0$,

$$I_{0+}^\alpha Q^p(y) = \frac{p}{\Gamma(\alpha + 1)} \int_0^{Q(y)} x^{p-1} (y - F(x))^\alpha dx, \quad (14.7)$$

while the Fabius-clock primitive is the elementary gamma law [equation \(4.12\)](#). These two answers are complementary: the first retains Euclidean probability time and has a Fabius kernel; the second uses the inverse clock and becomes algebraic.

15 Numerical experiments and reproducibility

The accompanying script `numerical_experiments.py` reconstructs up by FFT inversion of a 48-factor sinc product on a 2^{18} grid, builds monotone interpolants for F and Q , and compares formulas obtained from mathematically independent representations. The fractional-Laplacian profiles are then evaluated by the nonperiodic singular integral, not by a periodic Fourier multiplier, eliminating algebraic image contamination. Exact centered moments are generated by rational recurrence rather than by numerical quadrature. No random sampling is used.

15.1 Cross-check table

identity	direct side	transformed side	absolute error
Generalized Stieltjes moment series	0.466934130794	0.466934130794	1.665e-16
Generalized Stieltjes hierarchy	0.466934130794	0.466934130794	1.110e-16
Exterior Riesz moment series	-0.189818171242	-0.189818171242	7.272e-15
Caputo inverse transmutation	0.556844544942	0.556845008522	4.636e-07
Fractional F -up bridge	0.569952059655	0.56995167835	3.813e-07
Complex inverse resolvent	0.407510481292	0.407510481292	3.940e-13

The 10^{-7} -scale errors in the two Caputo rows are the discretization errors of an independent L1 finite-difference approximation. The transform and moment-series identities agree near machine precision.

15.2 Endpoint Caputo convergence

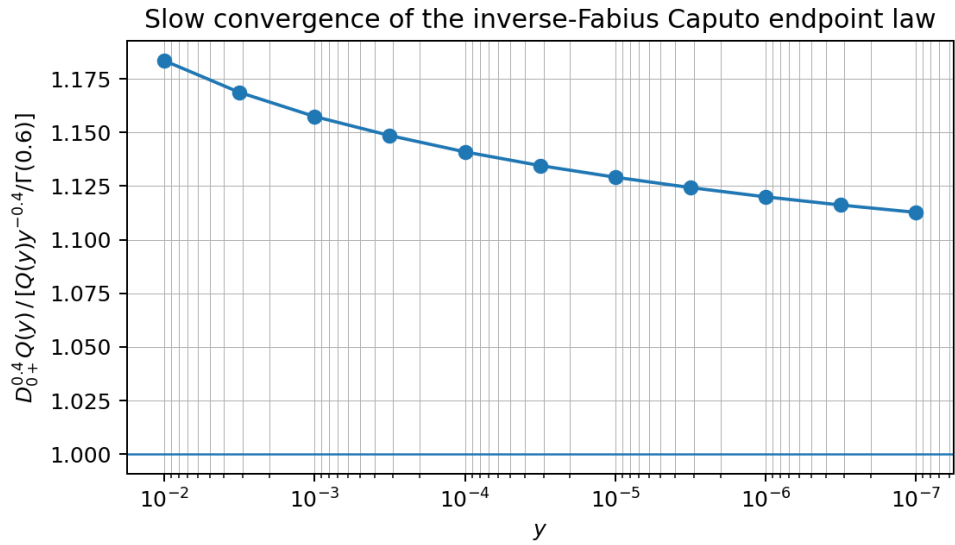


Figure 1: The ratio in equation (6.2) for $r = 1$ and $\beta = 0.4$. The theorem predicts convergence to 1; the slow drift exposes the scale of the first Lambert-type correction.

15.3 Fractional-Laplacian profiles

The script records all parameters and displayed numbers in `numerical_results.json`. Running

```
python numerical_experiments.py
```

regenerates both figures and the LaTeX table.

16 Conjectures and frontier directions

16.1 Phase-resolved Caputo endpoint correction

The proof of theorem 6.1 discards the narrow layer $t \approx 1$ where $F(qt)/F(q)$ is non-negligible. The width of that layer is controlled by the same saddle parameter that appears in the inverse Lambert- W expansion.

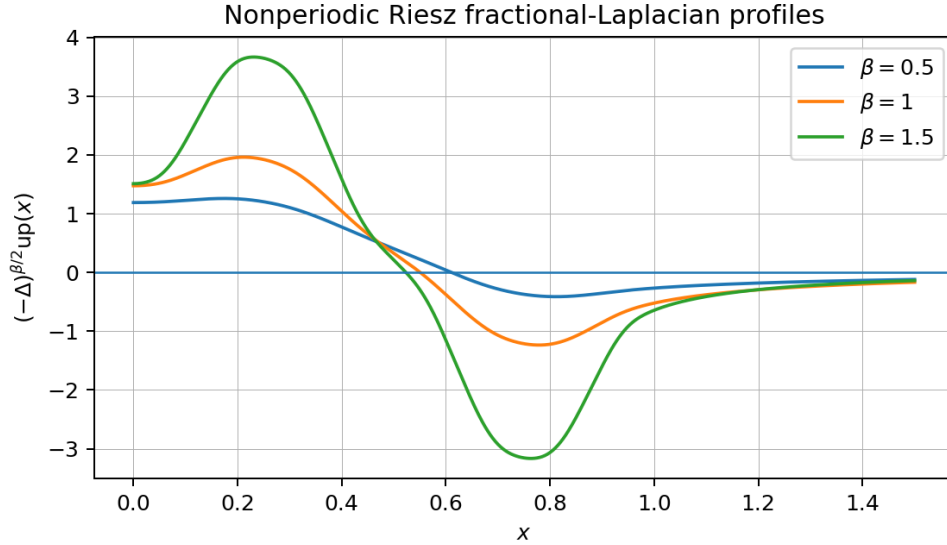


Figure 2: Nonperiodic singular-integral profiles of $(-\Delta)^{\beta/2} \text{up}$ for three representative orders. Each profile has one positive zero in the plotted half-axis, motivating Conjecture 11.4.

Conjecture 16.1 (Caputo endpoint phase). *For fixed $r > 0$ and $0 < \beta < 1$, there exists a bounded one-periodic function $A_{r,\beta}$ of the repository’s endpoint phase $\vartheta(y)$ such that*

$$\frac{{}^C D_{0+}^{\beta} Q(y)^r}{Q(y)^r y^{-\beta} / \Gamma(1-\beta)} = 1 + \frac{A_{r,\beta}(\vartheta(y))}{\rho(y)} + O(\rho(y)^{-2}), \quad (16.1)$$

where $\rho(y)$ is defined by equation (6.5). The periodic term should be nonconstant but extremely small, as in the endpoint density asymptotics.

A route to proof is a boundary-layer rescaling of equation (6.4), using the all-orders ratio asymptotic for $F(qt)/F(q)$ uniformly when $1 - t = O(\rho(y)^{-1})$.

16.2 The one-node law and total positivity

Conjecture 11.4 is stronger than the sign information proved in section 11. Three plausible approaches are:

1. represent the interior fractional Laplacian as a signed Stieltjes convolution and prove a variation-diminishing property;
2. use the infinite convolution construction of up and preservation of single crossing under uniform averaging;
3. establish strict total positivity for the appropriate fractional kernel after subtracting the center value.

A certified interval-arithmetic computation over a dense mesh in β would be a useful intermediate theorem, but not a substitute for uniqueness.

16.3 Fractional spectral interpolation

At the even integers,

$$\log_2 \frac{\mathcal{E}(2n)}{\mathcal{E}(0)} = n(n+1). \quad (16.2)$$

The explicit spectral measure from the derivative-norm report is a nonclassical representative of a shifted Stieltjes–Wigert moment problem. Determining the full Mellin transform $\mathcal{E}(s)$ would interpolate those quadratic samples and could reveal a small 2-periodic correction to the quadratic envelope

$$\frac{s^2}{4} + \frac{s}{2}. \quad (16.3)$$

The exact integer samples alone do not force such periodicity; this is an open spectral-analysis direction rather than a formal conjecture at present.

16.4 General base and q -deformation

Replace 2^{-j} by normalized b^{-j} weights. The clock conjugacy, inverse transmutation, generalized Stieltjes hierarchy, and exterior Riesz expansion survive with minimal change. The order-lowering factors become b -dependent, while the rational moment coefficients become $q = 1/b$ objects. The resulting family should provide a direct fractional-calculus bridge between the base- b random series and Gaussian q -binomial arithmetic.

16.5 Arithmetic of fractional constants

For rational β and algebraic $z > 1$, the values

$$S_\beta(z), \quad {}^C D_{0+}^\beta Q(y)^r, \quad \mathcal{E}(\beta), \quad a_\beta \quad (16.4)$$

are natural new constants. The Stieltjes series expresses the first through rational moments and gamma-polynomial coefficients. Padé approximation, Mahler-type functional equations, and denominator estimates from the q -binomial formulas may offer arithmetic information, although no irrationality claim is currently justified.

17 Lean formalization roadmap

The latest repository merge already formalizes the normalized integer Volterra operator and the exact Fabius primitive ladder. The new results can be layered on top in increasing order of analytic difficulty.

1. **Composition conjugacy.** Define $\mathcal{C}_F$ and $\mathcal{C}_Q$ on suitable function spaces; prove they are inverse order isomorphisms and transport semigroups.
2. **Fabius-clock integrals.** Formalize equation (4.6) first for positive real orders and continuous functions using the substitution theorem. The power law then reduces to Mathlib’s beta integral.
3. **First-order Caputo transmutation.** The proof of equation (5.3) requires only absolute continuity, inverse differentiation on $(0, 1)$, and a monotone change of variables. This is likely the most accessible genuinely new theorem.
4. **Endpoint asymptotics.** Package rapid variation and convexity into a dominated-convergence lemma for kernels $t^{r-1}(1 - F(qt)/F(q))^{-\beta}$.
5. **Generalized Stieltjes hierarchy.** Formalize the bounded random variable Z , complex powers for real $z > 1$, the moment series, and the smoothing equation. The order-lowering identity is then elementary.
6. **Exterior Riesz theorem.** Avoid principal values entirely by restricting first to $|x| > 1$; compact support reduces the proof to an ordinary integral and the Stieltjes identity.
7. **Weyl and full Riesz operators.** These require Fourier multiplier and tempered-distribution infrastructure and are a later target.

The numerical script can be converted into certified rational interval checks for selected transform identities and the conjectural zero locations, but the formal statements should remain independent of floating-point evidence.

18 Conclusion

The ordinary integral theory of the Fabius–Rvachev system is already unusually rigid: differentiation is conjugate to dyadic dilation, integer integration runs backward along the same orbit, and inverse substitution converts many areas into moments. The present investigation shows that this rigidity continues in several genuinely fractional directions.

The Fabius and inverse clocks give exact conjugate fractional calculi with semigroups, gamma power laws, and Mittag–Leffler eigenfunctions. In ordinary time, the inverse-composition transmutation removes the singular derivative of Q and replaces it by a Fabius-kernel integral; this leads to rigorous fractional endpoint laws. Standard fractional derivatives of F are exact fractional primitives of up, while whole-line Weyl derivatives satisfy a complex-order refinement hierarchy. On the transform side, one entire function $S_\lambda(z)$ unifies Appell polynomials, Cauchy resolvents, logarithmic potentials, Riesz tails, and fractional-Laplacian exterior fields.

The strongest next targets are now sharply posed: prove the one-node law, resolve the first phase-dependent correction to the inverse Caputo endpoint asymptotic, determine the continuous fractional-energy interpolation of the integer derivative norms, and formalize the first-order transmutation and Stieltjes hierarchy in Lean. None requires inventing a new ad hoc numerical object; each is anchored in an exact identity proved above.

A Formula index by operator

Operator or transform	Principal exact formula
Integer derivative of signed Fabius	$D^n \mathcal{F}(x) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n x)$
Integer primitive	$J^n \mathcal{F}(x) = 2^{\binom{n}{2}} \mathcal{F}(x/2^n)$
Weighted Volterra primitive	Newton series equation (3.4)
Inverse first primitive	$\int_0^y Q(v) dv = yQ(y) - F(Q(y)/2)$
Standard RL primitive of F	$I_{0+}^\alpha F(x) = \mathbb{E}[(x - Y)_+^\alpha] / \Gamma(\alpha + 1)$
Fabius-clock integral	kernel equation (4.6)
Fabius-clock power law	equations (4.8) and (4.9)
Caputo derivative of $H \circ Q$	transmutation equation (5.3)
Caputo derivative of Q^r	equation (5.5) and endpoint law equation (6.2)
Standard derivative of F	$D_{0+}^\beta F = 2^\beta I_{-1+}^{1-\beta} \text{up}(2x - 1)$ for $0 < \beta < 1$
Weyl derivative of up	complex-order hierarchy equation (8.5)
Generalized Stieltjes transform	moment series equation (9.2) and order lowering equation (9.5)
Complex F resolvent	equations (10.1) and (10.2)
Complex inverse resolvent	equation (10.6)
Exterior fractional Laplacian	$(-\Delta)^{\beta/2} \text{up} = -C_{1,\beta} S_{1+\beta}$
Exterior Riesz potential	$I_\alpha^R \text{up} = \kappa_\alpha S_{1-\alpha}$
Fractional energy	$\mathcal{E}(s) = (2\pi)^{-1} \int \xi ^s \phi ^2$

B Exact low-order moments

For convenience, the first one-sided moments are

$$m_0 = 1, \quad m_1 = \frac{1}{2}, \quad m_2 = \frac{5}{18}, \quad m_3 = \frac{1}{6}, \quad m_4 = \frac{143}{1350}, \quad m_5 = \frac{19}{270}. \quad (\text{B.1})$$

The centered moments in equation (2.14) satisfy

$$c_k = \frac{2m_{2k+1}}{2k+1}. \quad (\text{B.2})$$

Substitution in equation (9.2) produces exact rational-coefficient large- z expansions for every fixed complex order λ .

C Computational files

The archive accompanying this report contains:

- `fabius_fractional_integrals_report.tex` — the present source;
- `fabius_fractional_integrals_report.pdf` — the rendered article;
- `numerical_experiments.py` — commented reproducibility code;
- `numerical_results.json` and `numerical_results.txt` — machine- and human-readable output;
- `numerical_results.tex` — the generated table;
- `caputo_endpoint_ratio.png` and `riesz_fractional_profiles.png` — generated figures;
- `README.md` and `SHA256SUMS.txt`.

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